

● HARD

● EXPRESSION OF IDEAS

Expression of Ideas

10 questions · ~15 min

03

Q1

EXPRESSION OF IDEAS

A staunch supporter of women's voting rights, Wilhelmina Kekelaokalaninui Widemann Dowsett sought to coordinate the efforts of suffragists in her native Hawai'i. _____blank in 1912, she founded the National Women's Equal Suffrage Association of Hawai'i, an organization that lobbied for women's voting rights in the US territory.

Which choice completes the text with the most logical transition?

- A. In other words,
- B. Conversely,
- C. To that end,
- D. Alternatively,

While researching a topic, a student has taken the following notes:

- British musicians John Lennon and Paul McCartney shared writing credit for numerous Beatles songs.
- Many Lennon-McCartney songs were actually written by either Lennon or McCartney, not by both.
- The exact authorship of specific parts of many Beatles songs, such as the verse for “In My Life,” is disputed.
- Mark Glickman, Jason Brown, and Ryan Song used statistical methods to analyze the musical content of Beatles songs.
- They concluded that there is 18.9% probability that McCartney wrote the verse for “In My Life,” stating that the verse is “consistent with Lennon’s songwriting style.”

The student wants to make a generalization about the kind of study conducted by Glickman, Brown, and Song. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Based on statistical analysis, Glickman, Brown, and Song claim that John Lennon wrote the verse of “In My Life.”
- B. There is only an 18.9% probability that Paul McCartney wrote the verse for “In My Life”; John Lennon is the more likely author.
- C. It is likely that John Lennon, not Paul McCartney, wrote the verse for “In My Life.”
- D. Researchers have used statistical methods to address questions of authorship within the field of music.

In hindsight, given the ideas about the natural world circulating among British scientists in the 1800s, the theory of natural selection was an obvious next step. It may not have been a coincidence, _____ blank that Charles Darwin and Alfred Wallace arrived at the concept independently. Indeed, contrary to the popular myth of the lone genius, theirs is not the first paradigm-shifting theory to have emerged from multiple scholars working in parallel.

Which choice completes the text with the most logical transition?

- A. however,
- B. then,
- C. moreover,
- D. for example,

While researching a topic, a student has taken the following notes:

- A thermal inversion is a phenomenon where a layer of atmosphere is warmer than the layer beneath it.
- In 2022, a team of researchers studied the presence of thermal inversions in twenty-five gas giants.
- Gas giants are planets largely composed of helium and hydrogen.
- The team found that gas giants featuring a thermal inversion were also likely to contain heat-absorbing metals.
- One explanation for this relationship is that these metals may reside in a planet's upper atmosphere, where their absorbed heat causes an increase in temperature.

The student wants to present the study's findings to an audience already familiar with thermal inversions. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Heat-absorbing metals may reside in a planet's upper atmosphere.
- B. The team studied thermal inversions in twenty-five gas giants, which are largely composed of helium and hydrogen.
- C. Researchers found that gas giants featuring a thermal inversion were likely to contain heat-absorbing metals, which may reside in the planets' upper atmospheres.
- D. Gas giants were likely to contain heat-absorbing metals when they featured a layer of atmosphere warmer than the layer beneath it, researchers found; this phenomenon is known as a thermal inversion.

When one looks at the dark craggy vistas in Hitoshi Fugo's evocative photo series, one's mind might wander off to the cratered surfaces of faraway planets.

_____ blank it's the series' title, *Flying Frying Pan*, that brings one back to Earth, reminding the viewer that each photo is actually a close-up view of a familiar household object: a frying pan.

Which choice completes the text with the most logical transition?

- A. Consequently,
- B. Alternatively,
- C. Ultimately,
- D. Additionally,

While researching a topic, a student has taken the following notes:

- Cambodia's Angkor Wat was built in the 1100s to honor the Hindu god Vishnu.
- It has been a Buddhist temple since the sixteenth century.
- Decorrelation stretch analysis is a novel digital imaging technique that enhances the contrast between colors in a photograph.
- Archaeologist Noel Hidalgo Tan applied decorrelation stretch analysis to photographs he had taken of Angkor Wat's plaster walls.
- Tan's analysis revealed hundreds of images unknown to researchers.

The student wants to present Tan's research to an audience unfamiliar with Angkor Wat. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Tan photographed Angkor Wat's plaster walls and then applied decorrelation stretch analysis to the photographs.
- B. Decorrelation stretch analysis is a novel digital imaging technique that Tan used to enhance the contrast between colors in a photograph.
- C. Using a novel digital imaging technique, Tan revealed hundreds of images hidden on the walls of Angkor Wat, a Cambodian temple.
- D. Built to honor a Hindu god before becoming a Buddhist temple, Cambodia's Angkor Wat concealed hundreds of images on its plaster walls.

In a heated debate in biogeography, the field is divided between dispersalists and vicariancists. _____ blank there are those who argue that dispersal is the most crucial determining factor in a species' distribution, and those who insist that vicariance (separation due to geographic barriers) is. Biogeographer Isabel Sanmartín counts herself among neither.

Which choice completes the text with the most logical transition?

- A. Furthermore,
- B. By contrast,
- C. Similarly,
- D. That is,

While researching a topic, a student has taken the following notes:

- The factors that affect clutch size (the number of eggs laid at one time) have been well studied in birds but not in lizards.
- A team led by Shai Meiri of Tel Aviv University investigated which factors influence lizard clutch size.
- Meiri's team obtained clutch-size and habitat data for over 3,900 lizard species and analyzed the data with statistical models.
- Larger clutch size was associated with environments in higher latitudes that have more seasonal change.
- Lizards in higher-latitude environments may lay larger clutches to take advantage of shorter windows of favorable conditions.

The student wants to emphasize the aim of the research study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Researchers wanted to know which factors influence lizard egg clutch size because such factors have been well studied in birds but not in lizards.
- B. After they obtained data for over 3,900 lizard species, researchers determined that larger clutch size was associated with environments in higher latitudes that have more seasonal change.
- C. We now know that lizards in higher-latitude environments may lay larger clutches to take advantage of shorter windows of favorable conditions.
- D. Researchers obtained clutch-size and habitat data for over 3,900 lizard species and analyzed the data with statistical models.

In astrophysics, a ring of debris orbiting a larger object within the object's Roche limit is expected to persist as a ring, whereas a ring of debris orbiting outside this limit would likely accrete into a satellite (e.g., a moon). Bruno Morgado and colleagues, _____ blank detected a dense ring of material orbiting the trans-Neptunian object Quaoar at a distance of 2,500 miles, well outside the calculated Roche limit of 1,100 miles, that has remained intact.

Which choice completes the text with the most logical transition?

- A. though,
- B. for example,
- C. fittingly,
- D. likewise,

While researching a topic, a student has taken the following notes:

- Researchers in a 2021 study wanted to determine the rate at which 17 languages conveyed both information and syllables.
- They calculated the bits of information conveyed per second (the IR, or information rate).
- The IR was found to be approximately consistent across the 17 languages (an average of 39 bits per second).
- They calculated the number of syllables spoken per second (the SR, or syllable rate).
- Spanish had the second-fastest SR (7.7 syllables per second).
- Vietnamese had the sixteenth-fastest SR (5.3 syllables per second).

The student wants to present an overview of the study's findings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. The 2021 study determined the information rate (IR) of 17 languages in bits of information conveyed per second.
- B. Researchers found that information was conveyed more quickly in Spanish, at 7.7 syllables per second, than in Vietnamese, at 5.3 syllables per second.
- C. Vietnamese had the sixteenth-fastest syllable rate, lower than that of Spanish, which had the second-fastest; however, Spanish had the lower information rate of the two.
- D. Though some of the languages differed in number of syllables spoken per second, all 17 conveyed information at roughly the same rate.